

EECS 4312
Software Requirements Engineering
Summer 2025

Project Description
Requirement Verification for Software Systems Using Model Checking

Project Objective:

The goal of this team-based project is to **verify the accuracy** of the functional requirements of the **Smart Health & Wellness Center Automation** system, specifically focusing on the **scheduling & booking** and **payment & billing** features. You will use the requirements you already obtained for the system earlier in the lab assignments, extract requirements using **NLP tools**, apply **model checking** with **SPIN** and **LTL** properties, and ensure that the requirements are **consistent, correct**, and free from **conflicts**.

Project Tasks:

1. Requirement Extraction Using NLP Tools:

- **Task:** Extract **functional** and **non-functional requirements** from the **system description** that you already have (Lab 2).
- **Responsibilities:**
 - Use NLP tools (e.g., **spaCy** or **NLTK**, and **GPTs**) to extract relevant requirements.
 - Manually:
 - Verify the **completeness** of the requirements extracted using NLP tools (ensure no key requirements are missing – your reference will be the textual description from the lab and the solution guideline of Lab 3).
 - Ensure **consistency** and check for any **conflicts** within the extracted requirements.

2. Model Checking Using SPIN and LTL Properties:

- **Task:** Apply **SPIN** to verify the accuracy of the functional requirements of the **scheduling & booking** and **payment & billing** features by checking **LTL properties** on the models of the requirements.
- **Responsibilities:**
 - Write simple **LTL properties** to model correct system behaviors for **scheduling & booking** and **payment & billing**.
 - Run **model checking** using **SPIN** to identify potential **conflicts, inconsistencies, or violations** in the requirements.
 - Analyze **SPIN's** results and verify whether the requirements are consistent with the expected behavior of the system (the reference is the textual description from the lab).
 - **Example LTL Properties:**
 - **For Scheduling:** "A service can only be booked by one user at a time."
 - **For Payment:** "A user can only access the service after confirming payment."

- Suggest possible improvements or refinements to the requirements if conflicts or errors are found.

Deliverables:

1. **Report:**
 - Summary of the extracted **functional** requirements.
 - **Analysis** of the extracted requirements, focusing on **completeness, consistency, and correctness**.
 - **Model checking results**, including any detected **conflicts, inconsistencies, and violations**. A discussion of the implications of these results for the system's functional requirements.
 - **Role Descriptions**: Clearly specify the role of each team member in the project and describe their specific contributions to the tasks.
2. **Code and Scripts:**
 - **NLP Code/Scripts**: The scripts used to extract functional and non-functional requirements from the provided description.
 - **Model Checking Code**: The **SPIN** models along with the **LTL properties** used for verification.

Timeline:

- **Deadline**: The project is due by **July 28, 2025 @ 11:59pm**.
- **Interim Checkpoints**: Teams should have regular progress updates to ensure the tasks are on track (e.g., a brief weekly report or team meeting). A summary of the meeting outcomes (tasks and task assignment) should be submitted on eclass on the 18th and 25th of July.

Evaluation Criteria:

- **Accuracy of Requirement Extraction**: Correctness and completeness of the functional requirements extracted for scheduling and payment. (Optional: you can try to extract some nonfunctional requirements, but no need to use them in the verification process)
- **Effectiveness of Model Checking**: Proper use of **SPIN** and **LTL** to identify potential issues and validate the requirements.
- **Clarity of Analysis**: Well-documented analysis of conflicts, inconsistencies, and correctness based on the model checking results.
- **Team Collaboration**: Clear documentation of the **role of each team member** and their specific contributions to the project (through the progress reports).
- If your project submission has any doubts about its **quality**, you will be asked for an **oral interview to present your project work** and answer some questions about your solutions.

Evaluation Criteria	Description	Max Points
Analysis of AI Based Requirement Extraction	Detailed evaluation of the accuracy and completeness of the requirements extracted using AI.	25
Effectiveness of Model Checking	Proper application of SPIN and LTL properties to validate the AI generated functional requirements for scheduling & booking and payment & billing.	25
Clarity of Analysis of Model Checking	Clear documentation of the analysis of conflicts, inconsistencies, and correctness based on the SPIN model checking results.	25
Team Collaboration and Role Description	Clear documentation of the role of each team member and their specific contributions to the project (through the progress reports).	10
Documentation of Deliverables	Quality and completeness of the submitted report and code/scripts (NLP, SPIN, LTL).	15

Total Points: 100

Grading Scale:

- **90-100:** Excellent work, all criteria met with few or no issues.
- **80-89:** Good work, some minor issues or areas for improvement.
- **70-79:** Adequate work, several areas need improvement.
- **60-69:** Needs improvement in key areas, major issues in several parts.
- **Below 60:** Unacceptable or incomplete work.

This rubric ensures that students are evaluated on a comprehensive set of criteria, including technical execution (requirement extraction, model checking), analysis, collaboration, and documentation.

Criteria	Description	Partial Grading
Analysis of AI Based Requirement Extraction	Detailed evaluation of the accuracy and completeness of the requirements extracted using AI.	20–25 points: Thorough and insightful analysis of the AI-based requirement extraction process; clearly identifies strengths and weaknesses in the extraction, assesses coverage and accuracy critically, and supports findings with specific examples or evidence. No major omissions or inconsistencies in the extracted requirements.
		15–19 points: Good analysis that addresses the quality of the extraction and discusses most relevant aspects; some limitations in depth or detail. Extracted requirements are mostly accurate and complete, with only minor issues.
		10–14 points: Basic or superficial analysis with limited reflection on the extraction process; some key issues not identified. Several important requirements are missing or inconsistently presented.
		0–9 points: Minimal or no meaningful analysis of the extraction process; lacks understanding of what was done well or poorly. Extracted requirements are largely inaccurate, incomplete, or disorganized.
Effectiveness of Model Checking	Proper application of SPIN and LTL properties to validate the requirements for scheduling and payment.	20–25 points: Effective use of SPIN and LTL properties, identification of conflicts and violations, correct analysis of results.
		15–19 points: SPIN and LTL used correctly, but minor issues or incomplete validation of requirements.
		10–14 points: Some issues in applying SPIN or LTL properties, limited or incorrect validation of the requirements.
		0–9 points: Model checking poorly executed or not performed; requirements not verified against expected behaviors or violated properties not addressed.
Clarity of Analysis of Model Checking	Clear documentation of the analysis of conflicts, inconsistencies, and	20–25 points: Well-structured, clear analysis of the model checking results, explaining any detected issues and how they impact the requirements.

Criteria	Description	Partial Grading
	correctness based on the SPIN model checking results.	15-19 points: Analysis is mostly clear but lacks sufficient depth or has minor lapses in connecting results to the system's requirements. 10-14 points: Analysis unclear or incomplete, does not fully address the conflicts and their implications for system requirements. 0-9 points: Very poor or no analysis, does not address the key findings from the model checking process.
Team Collaboration and Role Description	Clear documentation of the role of each team member and their specific contributions to the project.	9-10 points: Excellent role distribution, each member's contributions are clearly outlined and well documented. 8-9 points: Reasonable role distribution, some members' contributions unclear or underrepresented. 6-7 points: Poor role distribution, unclear contributions, or overlaps. 0-6 points: No clear role documentation or major discrepancies in team contributions.
Documentation of Deliverables	Quality and completeness of the submitted report and code/scripts (NLP, SPIN, LTL).	12-15 points: Fully complete, well-documented report with clear references to the deliverables, including high-quality code/scripts for NLP and SPIN. 8-11 points: Complete report and code, but minor formatting or documentation issues. 4-7 points: Incomplete or unclear report/code, missing important sections or lacking necessary explanations. 0-3 points: No or very poor documentation and code.